

Notes: Bryzgalova, Pelger, & Zhu (JF, 2025)

Forest through the Trees: Building Cross-Sections of Stock Returns

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1 Big picture

- **Question.** How should researchers build characteristic-based test assets and tradable risk-factor building blocks from individual stocks?
- **Problem with conventional sorts.** Single, double, and triple sorts are easy to interpret, but they are crude unconditional partitions. They miss nonlinearities and interactions in characteristics, while large collections of sorts contain redundancy and noisy repackaging.
- **Core idea.** The paper proposes Asset Pricing Trees (AP Trees), which use decision-tree-style conditional splits to create managed portfolios, then prune the tree with a global asset-pricing objective.
- **Main objective.** The selected portfolios should span the stochastic discount factor (SDF) projected on individual stock returns, conditional on firm characteristics.
- **Why spanning matters.** If test assets do not span the SDF projection, an asset pricing model can price the chosen portfolios while still being badly misspecified for individual stocks.
- **Methodological innovation.** Pruning is not based on local prediction-error reduction. It is based on the global Sharpe ratio/SDF-spanning problem, estimated with robust mean-variance shrinkage and LASSO sparsity.
- **Empirical design.** The paper uses CRSP/Compustat monthly U.S. stock data from 1964 to 2016, builds portfolios from 10 firm characteristics, and evaluates performance out of sample.
- **Main result.** AP Trees produce low-dimensional, interpretable, long-only portfolios with much higher out-of-sample Sharpe ratios and larger alphas than traditional sorts or prediction-based ML portfolios.
- **Economic takeaway.** The key failure of standard test assets is omitted interaction structure. AP Trees recover more of the SDF using fewer, better diversified, and more interpretable portfolios.

2 Data and measurement

2.1 Characteristics and stocks

- The empirical sample uses CRSP/Compustat stocks from January 1964 to December 2016.
- The risk-free rate is the one-month Treasury bill rate.
- The paper builds 10 firm characteristics from standard accounting and market variables in the Kenneth French data library.
- Market-based characteristics are updated monthly.
- Accounting-based characteristics are updated at the end of June, following asset-pricing convention.
- AP Trees do not require each stock to have all 10 characteristics at all times. A tree based on a subset of characteristics uses only the data needed for that tree.

Interpretation: This is important because many machine-learning return-prediction methods require a balanced characteristic panel or imputation. AP Trees can preserve more stocks and avoid ad hoc missing-value rules.

2.2 Managed portfolios

The paper starts from individual excess returns $R_{t,i}$ and characteristics $C_{t-1,i}$. A managed portfolio is created by applying a basis function $f_j(C_{t-1,i})$ to individual stocks:

$$R_{t,j}^{man} = \sum_{i=1}^N f_j(C_{t-1,i}) R_{t,i}.$$

- In conventional sorts, $f_j(\cdot)$ is usually a quantile indicator.
- In AP Trees, $f_j(\cdot)$ is a region indicator generated by a sequence of conditional splits.
- Intermediate tree nodes and terminal tree nodes are both candidate managed portfolios.

Interpretation: Conventional sorting fixes a grid. AP Trees let the data choose which conditional regions matter for SDF spanning.

2.3 Training, validation, and testing

The empirical design divides the 53-year sample into three nonoverlapping blocks:

- **Training sample:** 1964:01 to 1983:12. Used to estimate means, covariances, and candidate portfolio weights for many tuning parameters.
- **Validation sample:** 1984:01 to 1993:12. Used to choose shrinkage and sparsity parameters.
- **Testing sample:** 1994:01 to 2016:12. Used only for out-of-sample evaluation.

Interpretation: The testing sample is not used to choose portfolios or tuning parameters. This is central because high-dimensional portfolio construction is vulnerable to in-sample overfitting.

2.4 Baseline cross-sections

- The baseline experiment considers triplets of characteristics: size plus two other characteristics.
- With 9 non-size characteristics, this gives 36 cross-sections.
- Baseline AP Trees use depth four and select 40 portfolios.
- The main benchmarks are 32 and 64 triple-sorted portfolios.
- All portfolios are value-weighted.
- Level-four nodes based on repeated splits of only one characteristic are excluded to avoid giving AP Trees an unfair tail-sorting advantage.

Interpretation: The comparison is deliberately close to familiar Fama-French-style sorting. The difference is that AP Trees allow conditional split order and pruning rather than a fixed unconditional grid.

3 Models and empirical specifications

3.1 Conditional SDF projected on characteristics

Let R_t be the vector of individual stock excess returns. The SDF projected on characteristics is written as

$$M_t^C = 1 - \sum_{i=1}^N b_{t-1,i} (R_{t,i} - E_{t-1}[R_{t,i}]), \quad b_{t-1,i} = f(C_{t-1,i}).$$

The function $f(\cdot)$ can be nonlinear and nonseparable across characteristics.

Interpretation: The target is not merely conditional expected returns. The target is the SDF projection, which depends on the mean-variance structure of all characteristic-based returns.

3.2 From individual stocks to managed portfolios

Approximate the SDF loading function by basis functions:

$$f(C_{t-1,i}) \approx \sum_{j=1}^J f_j(C_{t-1,i}) w_j.$$

Then the conditional SDF can be written as an unconditional model on managed portfolios:

$$M_t^C = 1 - \sum_{j=1}^J w_j (R_{t,j}^{man} - E[R_{t,j}^{man}]), \quad R_{t,j}^{man} = \sum_{i=1}^N f_j(C_{t-1,i}) R_{t,i}.$$

Interpretation: This translation lets conventional unconditional asset-pricing tools be used, but only if the managed portfolios are rich enough to span the conditional SDF.

3.3 Omitted test assets

Suppose the SDF requires selected and omitted basis functions:

$$f(C_{t-1,i}) = \sum_{j=1}^{J_{select}} f_j^{select}(C_{t-1,i}) w_j^{select} + \sum_{j=1}^{J_{omit}} f_j^{omit}(C_{t-1,i}) w_j^{omit}.$$

A researcher prices only R_t^{select} with a traded factor model F_t :

$$R_t^{select} = \alpha^{select} + \beta^{select\top} F_t + \epsilon_t^{select}, \quad \alpha^{select} = 0.$$

The same factors may still misprice omitted assets:

$$R_t^{omit} = \alpha^{omit} + \beta^{omit\top} F_t + \epsilon_t^{omit}, \quad \Sigma^{omit} = \text{Var}(\epsilon_t^{omit}).$$

Interpretation: Even perfect pricing of a chosen cross-section is not enough. The cross-section must itself contain the relevant investment opportunities.

3.4 Proposition 1: misspecification due to omitted test assets

Let $SR(R)$ denote the maximum Sharpe ratio achievable from assets R . If F prices R^{select} , then

$$SR^2(R^{select}, R^{omit}) - SR^2(F) \leq \alpha^{omit\top} (\Sigma^{omit})^{-1} \alpha^{omit} \leq SR^2(R^{select}, R^{omit}) - SR^2(R^{select}).$$

Interpretation: The gap between the Sharpe ratio of the full characteristic-managed space and the chosen test assets bounds how badly omitted assets can be mispriced. This is the formal reason to compare the Sharpe ratios of AP Trees and conventional sorts.

3.5 AP Trees as conditional basis functions

A tree of depth d recursively splits stocks by characteristics. With M sorting variables, the set of possible split orders generates

$$M^d \cdot 2^d$$

overlapping portfolios when final and intermediate nodes are included.

- The regions are conditional: later splits depend on earlier splits.
- A node is a long-only portfolio of stocks in that region.
- Intermediate nodes are broader, more diversified portfolios.
- Terminal nodes are more granular, less diversified portfolios.

Interpretation: AP Trees approximate nonlinear and interaction effects while retaining interpretability because every portfolio can be traced to a path of characteristic splits.

3.6 Pruning as sparse SDF recovery

Let $\hat{\mu}$ and $\hat{\Sigma}$ be sample mean and covariance estimates for all AP Tree candidate portfolios. The population SDF weights solve

$$\omega = \Sigma^{-1}\mu.$$

The naive sample estimate overfits. AP Pruning replaces sample moments by robust moments and imposes sparsity:

$$\min_{\omega} \frac{1}{2}(\hat{\mu}^{robust} - \hat{\Sigma}^{robust}\omega)^{\top}(\hat{\Sigma}^{robust})^{-1}(\hat{\mu}^{robust} - \hat{\Sigma}^{robust}\omega) + \lambda_1\|\omega\|_1,$$

where

$$\hat{\Sigma}^{robust} = \hat{\Sigma} + \lambda_2 I_N, \quad \hat{\mu}^{robust} = \hat{\mu} + \lambda_0 \mathbf{1}.$$

Interpretation: λ_1 selects a sparse set of portfolios, λ_2 regularizes covariance estimation, and λ_0 shrinks means toward a common level.

3.7 Equivalent robust tangency portfolio

The paper shows that AP Pruning is equivalent to robust tangency portfolio construction:

$$\min_w \frac{1}{2}w^{\top}\hat{\Sigma}w + \lambda_1\|w\|_1 + \frac{1}{2}\lambda_2\|w\|_2^2$$

subject to

$$w^{\top}\mathbf{1} = 1, \quad w^{\top}(\hat{\mu} + r_f\mathbf{1}) \geq \mu_0 + r_f.$$

Interpretation: The target return μ_0 corresponds to mean shrinkage; the ridge penalty corresponds to variance shrinkage. LASSO controls how many basis portfolios remain.

3.8 Robust SDF recovery

For the active set of nonzero weights,

$$\left[(\hat{\Sigma} + \lambda_2 I_N)\hat{\omega}^{robust}\right]_i = \hat{\mu}_i + \lambda_0 - \lambda_1 \text{sign}(\hat{\omega}_i^{robust}).$$

For a diagonalized return rotation, the robust weights simplify to

$$\hat{\omega}^{robust} = (\hat{\Sigma} + \lambda_2 I_N)^{-1}(\hat{\mu} - \lambda_1 \mathbf{1})_+.$$

Interpretation: The method generalizes robust SDF recovery by combining mean shrinkage, covariance shrinkage, and sparsity.

3.9 Cross-sectional pricing metric

The paper reports the uncentered cross-sectional R^2 :

$$XS-R^2 = 1 - \frac{N}{N-K} \frac{\sum_{i=1}^N \alpha_i^2}{\sum_{i=1}^N E[R_i^{ex}]^2}.$$

Interpretation: Low $XS-R^2$ means large remaining pricing errors relative to the level of expected returns. AP Tree portfolios are often harder for standard factors to price, which is evidence that they reveal omitted SDF information.

4 Identification and empirical design

4.1 What identifies the comparison

- The comparison holds the same underlying characteristics fixed.
- AP Trees, triple sorts, cross-section-specific factors, V-Trees, and ML prediction portfolios are all judged by out-of-sample SDF performance.
- The test sample is not used for portfolio selection or tuning.
- Mean and variance shrinkage are optimized in validation for every cross-section, not just AP Trees.

Interpretation: The design isolates the construction of test assets. The same characteristic information can be packaged well or poorly.

4.2 Why local tree pruning is not enough

- Standard decision trees split nodes using local criteria, usually prediction error.
- SDF spanning is global: a split is useful only if it improves the risk-return trade-off after accounting for all other candidate portfolios.
- A child node can have the same mean as its parent but still matter because of covariance with other nodes.

Interpretation: AP Trees are not just random forests applied to returns. The pruning objective is an asset-pricing objective.

4.3 Why out-of-sample Sharpe ratios matter

- In-sample maximum Sharpe ratios can be artificially high in many-asset problems.
- The paper uses training/validation/testing splits and robust shrinkage to avoid overstating performance.
- The out-of-sample Sharpe ratio measures how much SDF variation is actually captured by the chosen cross-section.

Interpretation: A high out-of-sample SR for a small set of test assets implies that the portfolios retain economically relevant information rather than merely overfitting.

4.4 What the paper cannot fully prove

- AP Trees approximate the SDF projection for a chosen set of characteristics; they do not prove that those characteristics exhaust all pricing information.
- The selected portfolios depend on tuning choices, sample splits, and the candidate characteristic set.
- The paper evaluates characteristic-based cross-sections rather than estimating a structural economic model of why the selected regions carry premia.
- The approach delivers better test assets and factor building blocks, but it does not by itself identify the economic source of every premium.

Interpretation: The contribution is methodological and empirical: better characteristic-based portfolios for spanning the projected SDF.

5 Tables and figures

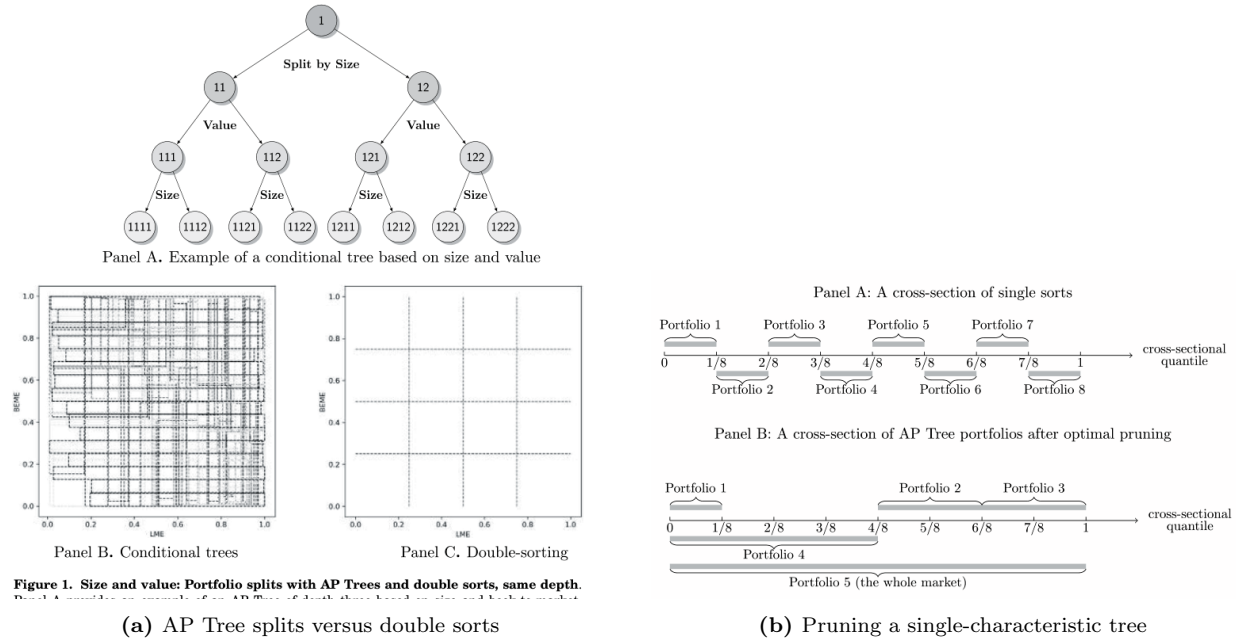
The visuals below are grouped to save space. They are directly cropped from the paper.

5.1 Figure 1 + Figure 2: AP Trees versus conventional sorts and pruning

Read:

- Figure 1 shows that AP Trees create conditional partitions: the same characteristic can be used at different depths and after different prior splits.
- Double sorting imposes a fixed rectangular grid, while AP Trees generate many possible conditional regions.
- Figure 2 shows that pruning need not keep equal-sized quantile bins. It can merge nearby regions when the split does not improve SDF spanning.

Economic message: AP Trees keep the interpretability of sorting but avoid the rigidity of a fixed grid.



5.2 Figure 3 + Figure 4: Global pruning and equivalent tree representations

Read:

- Figure 3 illustrates pruning with multiple characteristics. Some intermediate nodes remain, while only some branches are split further.
- Figure 4 shows that different ways of representing a selected tree can span the same basis functions.
- The recursive identity is $f_1 = f_{11} + f_{12}$, so selecting a parent plus one child can span the same functions as selecting both children.

Economic message: The object that matters is the span of selected basis functions, not a literal final-node-only tree.

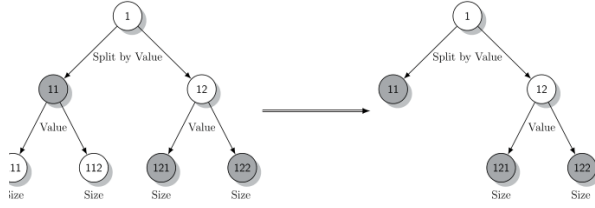


Figure 3. Illustration of pruning with multiple characteristics. The figure shows a sample

(a) Pruning with multiple characteristics

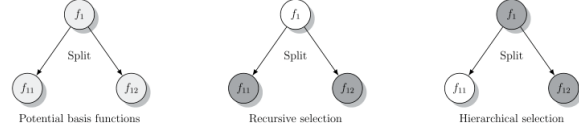


Figure 4. Equivalence of different representations of a tree. The figure shows two equivalent ways of selecting nodes in a split. Dark-colored nodes are selected, while white ones are

(b) Equivalent representations of a tree

5.3 Figure 5 + Figure 6: Empirical design and baseline three-characteristic results

Read:

- Figure 5 documents the training, validation, and testing split: 20 years, 10 years, and 23 years.
- Figure 6 compares AP Trees, 32/64 triple sorts, and cross-section-specific factors across 36 three-characteristic cross-sections.
- AP Trees generally deliver higher out-of-sample Sharpe ratios.
- AP Tree SDF alphas relative to FF5 tend to have larger t -statistics.
- The $XS-R^2$ panels show that AP Tree cross-sections are harder for the FF5 model to span.

Economic message: AP Trees reveal more pricing information from the same characteristics than standard triple sorts.

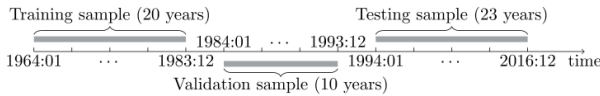
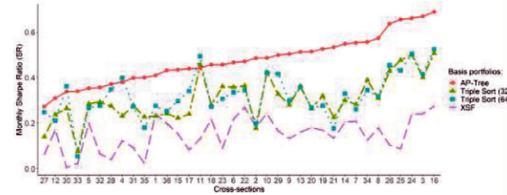
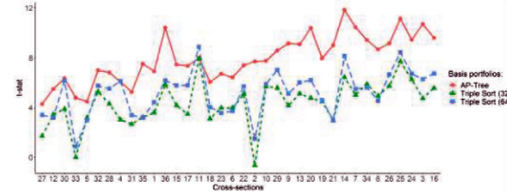


Figure 5. Timeline of the empirical strategy. The SDF weights and portfolio components are estimated on the training data (first 20 years). The shrinkage parameters are chosen based on

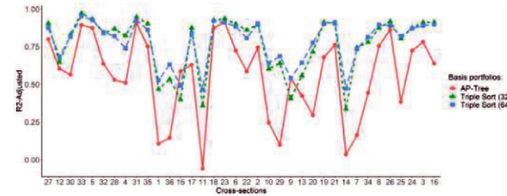
(a) Training-validation-testing timeline



Panel A. Sharpe ratio of the robust SDF spanned by AP-Trees and conventional sorts.



Panel B. Alpha of the robust SDF spanned by AP-Trees and conventional sorts w.r.t. $FF5$.



Panel C. Cross-sectional R^2 with AP-Trees and triple sorts as basis assets w.r.t. $FF5$.

(b) AP Trees versus triple sorts across 36 cross-sections

5.4 Figure 7 + Figure 8: How many portfolios and the role of interactions

Read:

- Figure 7 shows that using only 10 AP Tree portfolios retains much of the performance of larger AP Tree cross-sections.
- Figure 8 removes interaction nodes and shows a large decline in Sharpe ratios.
- The no-interaction AP Tree still contains nonlinear single-characteristic information, but loses much of the conditional interaction channel.

Economic message: Interactions across characteristics are a main reason AP Trees outperform conventional sorts.

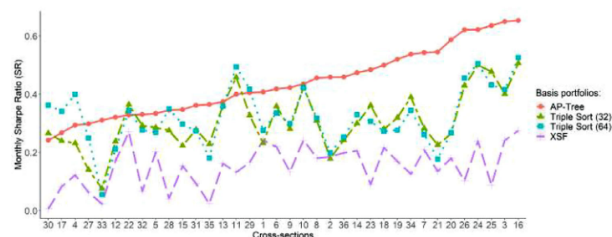


Figure 7. AP Trees (10 portfolios) versus triple sorts. The figure displays monthly out-of-sample Sharpe ratios of the robust MVE portfolios spanned by AP Trees, pruned to 10 portfolios, relative to those spanned by 32/64 triple sorts and cross-section-specific factors (XSF). All of the cross-sections are sorted by the SR achieved with AP Trees pruned to 10 portfolios.

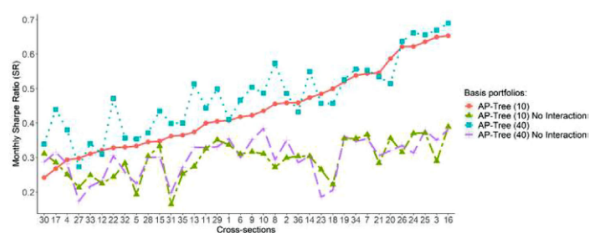


Figure 8. AP Trees: The impact of interaction nodes. The figure displays monthly out-of-sample SRs of the robust MVE portfolios spanned by AP Trees with and without interaction nodes. All of the cross-sections are sorted by the Sharpe ratio achieved with AP Trees pruned to 10 portfolios.

We use a LASSO model to build cross-sections of different size and find

(a) AP Trees with 10 portfolios versus triple sorts

7. Nonlinearities and Characteristic Interactions

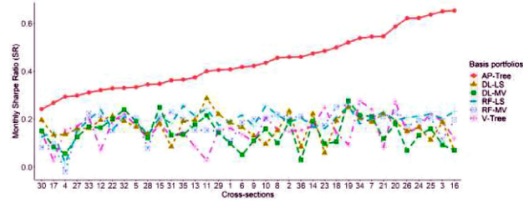
(b) Impact of interaction nodes

5.5 Figure 9 + Table I: AP Trees versus ML portfolios and a detailed benchmark

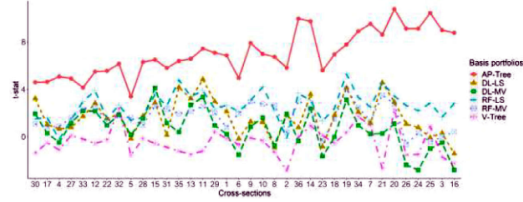
Read:

- Figure 9 compares AP Trees with V-Trees and machine-learning prediction portfolios built from deep learning and random forests.
- AP Trees have higher out-of-sample Sharpe ratios and larger FF5 alphas.
- Table I focuses on size, operating profitability, and investment.
- AP Trees (10) have SR 0.65 and AP Trees (40) have SR 0.69, versus 0.51 and 0.53 for 32/64 triple sorts.
- FF5 alpha t -statistics are 8.76 and 9.60 for AP Trees, versus 5.57 and 6.73 for triple sorts.

Economic message: Prediction accuracy alone is not the same as SDF spanning. AP Trees target the asset-pricing object directly.



Panel A. SRs of robust SDFs spanned by AP Trees and standard ML-based portfolios.



Panel B. Alphas of robust SDFs spanned by AP Trees and standard ML-based portfolios w.r.t. $FF5$.

(a) AP Trees versus off-the-shelf ML portfolios

Table I
Cross-Sections Based on Size, Operating Profitability, and Investment

The table presents aggregate properties of the cross-sections based on size, investment, and operating profitability, created from AP Trees (pruned to 10 and 40 portfolios) and triple sorts (32 and 64 portfolios). For each cross-section, the table reports its monthly SR on the test sample, along with the alpha of the SDF spanned by the corresponding basis portfolios. Alphas are computed relative to the Fama-French three- and five-factor models, cross-section-specific factors (market, size, investment, and operating profitability), and the composite FF11 model.

		Type of Cross-Section			
		AP Trees (10)	AP Trees (40)	TS (32)	TS(64)
SDF SR		0.65	0.69	0.51	0.53
	FF3	0.94	0.90	0.75	0.84
		[10.11]	[11.03]	[7.40]	[8.13]
	FF5	0.81	0.76	0.47	0.61
		[8.76]	[9.60]	[5.57]	[6.73]
	XSF	0.81	0.76	0.46	0.61
$XS-R^2$		[8.77]	[9.46]	[5.39]	[6.69]
	FF11	0.89	0.80	0.37	0.65
		[9.12]	[9.60]	[4.29]	[6.91]
	FF3	18.0%	51.0%	82.0%	82.0%
	FF5	11.0%	64.0%	91.0%	90.0%
	XSF	28.0%	65.0%	91.0%	90.0%
	FF11	—	42.0%	92.0%	87.0%

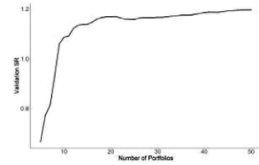
(b) Cross-sections based on size, OP, and investment

5.6 Figure 10 + Table II: Tuning and selected AP Tree portfolios

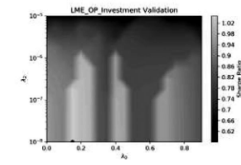
Read:

- Figure 10 shows validation and testing Sharpe ratios as functions of tuning parameters.
- Validation selects the number of portfolios and the mean/variance shrinkage parameters.
- Table II shows that the selected AP Tree portfolios are interpretable: market, medium caps, low profitability, low-investment small caps, high-investment small caps, and related conditional regions.
- Some selected portfolios are intermediate nodes, not just terminal nodes.

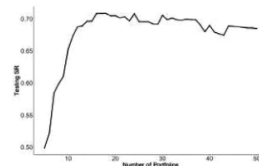
Economic message: The selected test assets are not arbitrary black-box portfolios. They can be described as conditional characteristic groups.



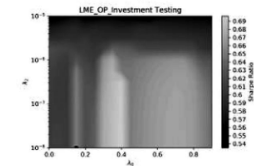
Panel A. Validation: Portfolio number impact.



Panel B. Validation: Mean and variance shrinkage.



Panel C. Testing: Portfolio number impact.



Panel D. Testing: Mean and variance shrinkage.

(a) Tuning parameters and out-of-sample performance

(b) Selected AP Tree portfolios

5.7 Table II continued + Figure 11: Triple-sort challenging portfolios and AP Tree regions

Read:

Table II
Portfolios in the Cross-Sections

The table presents properties of the portfolios, spanning the impact of size, investment, and operating profitability on asset returns. Panel A presents the set of 10 portfolios. Created from trees of depth four (which exclude the extreme portfolios, representing 1/6 of the stocks sorted by the same characteristic) and their features: the average percentage of currently available stocks, included in the portfolio, their value-weighted quantile based on size, and alphas with respect to Fama-French three- and five-factor models, the model that includes cross-section-specific factors in addition to the market, and the FF11 model (market and 10 long-short portfolios based on all available characteristics), with corresponding t -stats in the brackets. *, **, and *** correspond to 10%, 5%, and 1% significance levels, respectively. Panel B presents the same statistics for the 10 portfolios out of 64, created by triple-sorting, that are the most challenging to price, according to the composite FF11 model. The portfolio ID for AP Trees denotes the sequence of characteristics (LME/OP/INV) in the first part and the direction (low/high) of the split in the second part. Figure 11 illustrates the composition for all 10 AP Tree portfolios. The portfolio ID for triple sorts denotes the quantile for (LME/OP/INV).

Portfolio ID	Portfolio Description	% of stocks	WV Size Quantile	α_{FF3}	α_{FF5}	α_{XSF}	α_{FF11}
Panel A: 10 AP-Tree Portfolios							
1 (1111.1)	Market	100%	0.95	—	—	—	—
2 (1111.121)	Medium caps	25%	0.71	[-0.34]	[-0.36]	[-0.40]	[-0.26]
3 (3111.11)	Low profitability	50%	0.94	0.03	-0.11**	-0.11**	-0.11**
4 (3111.1111)	Low investment, small caps	6.25%	0.07	1.49***	1.32***	1.32***	1.63***
5 (3211.1121)	Low investment, high profit., small caps	12.5%	0.51	[6.14]	[5.33]	[5.33]	[6.47]
6 (1221.1111)	Small caps, low profitability	12.5%	0.37	-0.37***	-0.34***	-0.34***	-0.29**
7 (2221.1111)	Low profitability, small caps	6.25%	0.19	[-2.81]	[-3.00]	[-3.01]	[-2.41]
8 (3331.1221)	High investment, small caps	6.25%	0.45	[-0.717]	[-0.95]	[-0.89]	[0.21]
9 (2122.1111)	Low profitability, small caps	6.25%	0.27	[-5.86]	[-4.81]	[-4.78]	[-3.75]
10 (3131.1212)	High investment, small caps	6.25%	0.48	[-0.37]	[-0.42]	[-0.42]	[-0.30]
				[-1.95]	[-2.21]	[-2.13]	[-1.00]
				[-0.84]	[-0.64]	[-0.64]	[-0.50]
				[-5.92]	[-4.84]	[-4.81]	[-3.92]

(Continued)

- The triple-sort portfolios that are hardest to price tend to be narrow cells containing about 1% to 5% of stocks.
- AP Tree portfolios often use larger conditional regions, which improves diversification.
- Figure 11 maps the selected AP Tree portfolios in the size-profitability-investment characteristic cube.

Economic message: AP Trees often capture the same pricing information with broader and more stable regions than triple sorts.

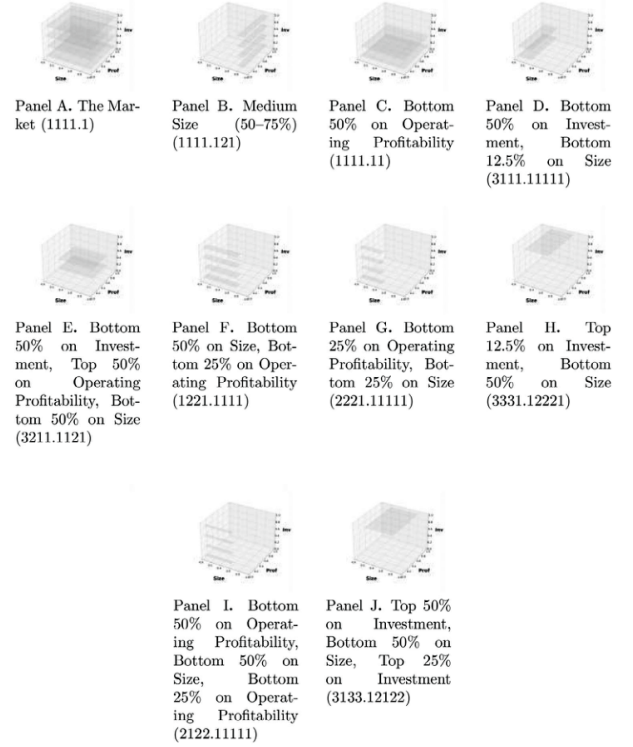


Figure 11. AP Tree (10) portfolios for size, investment, and operating profitability. The panels highlight the composition of the 10 AP Tree portfolios in the characteristic space. The tree

Table II—Continued

Panel B: Top 10 Most “Challenging” Assets from 64 Triple-Sorted Portfolios									
122	LME(0 to 0.25) × OP(0.25 to 0.5) × Inv(0.25 to 0.5)	1.91 %	0.17	0.97***	0.84***	0.84***	0.95***		
				(6.61)	(5.60)	(5.54)	(6.61)		
214	LME(0.25 to 0.5) × OP(0 to 0.25) × Inv(0.75 to 1)	1.55 %	0.40	−0.93***	−0.61***	−0.61***	−0.66***		
				[−5.27]	[−3.60]	[−3.61]	[−3.77]		
132	LME(0 to 0.25) × OP(0.5 to 0.75) × Inv(0.25 to 0.5)	1.03 %	0.17	0.39*	0.40*	0.38*	0.77***		
				(1.83)	(1.90)	(1.74)	(3.65)		
121	LME(0 to 0.25) × OP(0.25 to 0.5) × Inv(0 to 0.25)	1.9 %	0.17	0.59***	0.45**	0.44**	0.66***		
				(3.38)	(2.48)	(2.45)	(3.56)		
112	LME(0 to 0.25) × OP(0 to 0.25) × Inv(0.25 to 0.5)	1.66 %	0.17	0.24	0.25	0.24	0.49***		
				(1.45)	(1.45)	(1.38)	(2.79)		
123	LME(0 to 0.25) × OP(0.25 to 0.5) × Inv(0.5 to 0.75)	1.47 %	0.17	0.38**	0.32**	0.32**	0.43***		
				(2.56)	(2.10)	(2.08)	(2.71)		
222	LME(0.25 to 0.5) × OP(0.25 to 0.5) × Inv(0.25 to 0.5)	1.95 %	0.40	0.46***	0.35***	0.34***	0.28**		
				(4.09)	(3.52)	(2.98)	(2.69)		
244	LME(0.25 to 0.5) × OP(0.75 to 1) × Inv(0.75 to 1)	1.18 %	0.41	−0.57***	−0.47***	−0.49**	−0.34**		
				[−3.44]	[−3.18]	[−2.65]	[−2.28]		
334	LME(0.5 to 0.75) × OP(0.5 to 0.75) × Inv(0.75 to 1)	1.08 %	0.66	0.35**	0.34**	0.32*	0.37**		
				(2.33)	(2.18)	(1.77)	(2.25)		
111	LME(0 to 0.25) × OP(0 to 0.25) × Inv(0 to 0.25)	5.04 %	0.16	0.42*	0.21	0.22	0.48**		
				(1.89)	(0.95)	(0.97)	(2.19)		

(a) Top challenging triple-sort assets

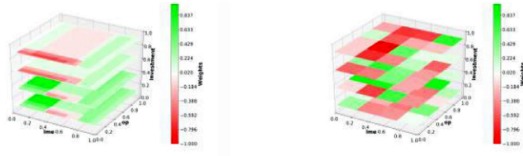
(b) Composition of 10 AP Tree portfolios

5.8 Figure 12 + Figure 13: SDF weights and time variation

Read:

- Figure 12 compares the conditional SDF weights implied by 10 AP Tree portfolios and 64 triple-sorted portfolios.
- AP Trees reveal a smoother but richer conditional structure than the coarse triple-sort grid.
- Figure 13 compares constant and rolling 20-year SDF weights.
- Time-varying weights do not overturn the central result; AP Tree portfolios continue to span higher-SR SDFs.

Economic message: The AP Tree advantage is not just a static-weight artifact. It persists when SDF weights are estimated over rolling windows.



Panel A. SDF spanned by AP Trees (10) Panel B. SDF spanned by TS (64)

Figure 12. SDF weights in AP Trees and triple sorts. These panels show the structure of the conditional SDF spanned by 10 AP Trees portfolios and 64 triple-sorted basis assets for size, investment, and operating profitability.

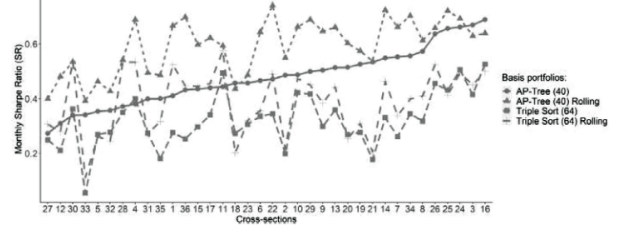


Figure 13. SRs of the SDF estimated on a rolling window. The figure displays the monthly out-of-sample SR of the robust MVE portfolios with time-varying weights estimated on a rolling window of 20 years and constant weights. The basis portfolios are the pruned AP Trees (40 and 10 portfolios) and triple sorts. The AP Tree portfolios are selected on the training data and kept

(a) SDF weights in AP Trees and triple sorts

(b) Rolling-window SDF Sharpe ratios

5.9 Table III: Small-cap robustness

Read:

- Panel A removes portfolios whose value-weighted size quantile is below 0.4.
- Panel B restricts the universe to stocks with market capitalization above 0.01% of total market capitalization.
- AP Trees retain stronger performance than triple sorts in both restrictions.
- For example, in Panel B, AP Trees (10) have SR 0.30 versus 0.17 for TS(32) and TS(64).

Economic message: The AP Tree advantage is not solely a microcap effect.

Table III
Size, Operating Profitability, and Investment: Cross-Sections without Small Caps

The table presents the aggregate properties of the cross-sections based on size, operating profitability, and investment, created from AP Trees (pruned to 10 and 40 portfolios) and triple sorts (32 and 64 portfolios). Panel A is based only on the basis portfolios that have a value-weighted size quantile greater than 0.4, while Panel B describes cross-sections created only from stocks with market capitalization greater than 0.01% of the total market capitalization (currently, around top 600 stocks in the United States). For each cross-section, the table reports its monthly SR on the test sample, along with the alpha of the SDF spanned by the corresponding basis portfolios. Alphas are computed relative to the Fama-French three- and five-factor models, cross-section-specific factors (market and long-short portfolios, reflecting size, operating profitability, and investment), and the composite FF11 model, which includes the market portfolio and the 10 long-short portfolios, based on the cross-sectional characteristics.

		Panel A: Portfolios with Size Quantile above 0.4				Panel B: Stocks with Market Cap above 0.01%			
		AP-Trees (10)	AP-Trees (40)	TS (32)	TS(64)	AP-Trees (10)	AP-Trees (40)	TS (32)	TS(64)
SDF SR	FF3	0.48	0.45	0.25	0.40	0.30	0.22	0.17	0.17
	FF5	0.68	0.73	0.47	0.76	0.70	0.89	0.77	0.82
	FF11	[7.51]	[7.64]	[2.30]	[6.00]	[5.04]	[4.27]	[2.74]	[2.78]
	XSF	0.45	0.47	−0.11	0.54	0.32	0.37	0.16	0.25
	FF11	[5.70]	[6.11]	[−0.72]	[4.44]	[3.87]	[2.57]	[0.72]	[1.00]
XS-R ²	FF3	0.44	0.47	−0.11	0.54	0.31	0.36	0.14	0.22
	FF5	[4.76]	[5.04]	[−0.74]	[4.21]	[3.75]	[2.49]	[0.65]	[0.88]
	FF11	0.43	0.43	−0.25	0.34	0.29	0.36	0.18	0.38
	XSF	[5.37]	[5.27]	[−1.62]	[2.76]	[3.71]	[2.45]	[0.83]	[1.55]
	FF11	62.0%	73.0%	76.0%	86.0%	42.0%	96.0%	91.0%	90.0%
	XSF	70.0%	85.0%	91.0%	93.0%	70.0%	98.0%	95.0%	93.0%
	FF11	76.0%	85.0%	92.0%	93.0%	71.0%	98.0%	95.0%	93.0%
	FF11	—	86.0%	84.0%	94.0%	—	98.0%	95.0%	93.0%

5.10 Figure 14 + Figure 15: Large-dimensional cross-sections

Read:

- Figure 14 studies 10 characteristics and compares AP Trees with quintile sorts, decile sorts, and size-based double-sort collections.
- AP Trees deliver higher out-of-sample SRs with a relatively small number of selected portfolios.
- Figure 15 compares AP Trees with V-Trees and prediction-based portfolios.
- V-Trees and prediction portfolios have substantially lower SRs than AP Trees.

Economic message: In large characteristic spaces, the main value is not more portfolios. It is a better conditional grouping that targets SDF spanning.

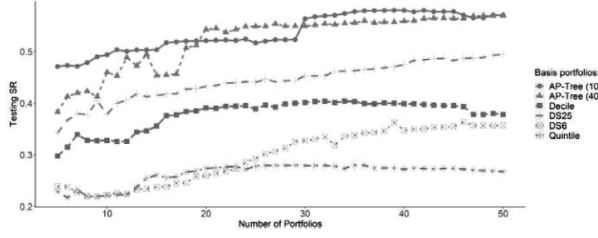


Figure 14. SRs of the SDFs spanned by AP Trees and traditional sorts. This figure shows out-of-sample SRs of SDFs based on 10 characteristics as a function of the number of basis assets constructed with AP Trees (based on 36 AP-Trees with 10 or 40 portfolios in each), 10 quintile sorts,

(a) AP Trees versus traditional large-dimensional sorts

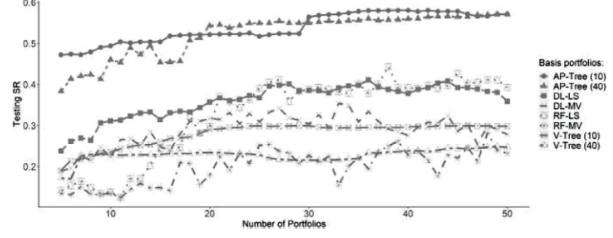


Figure 15. SR of the SDFs spanned by AP-Trees, V-Trees, and prediction portfolios. Out-of-sample Sharpe ratios of SDFs based on 10 characteristics as a function of the number of basis assets constructed with AP-Trees (based on 36 AP-Trees with 10, respectively, and 40 portfolios in

(b) AP Trees versus V-Trees and prediction portfolios

6 Short summary

- The paper asks how to construct cross-sections of stock returns that are useful as test assets and factor-building blocks.
- Traditional sorts are transparent but miss nonlinear and interaction effects.
- The paper reframes test-asset construction as a problem of spanning the SDF projected on individual stocks conditional on characteristics.
- AP Trees generate many conditional long-only portfolios from decision-tree splits.
- AP Pruning selects a sparse subset of final and intermediate nodes using a global SDF-spanning objective.
- Robust mean-variance shrinkage controls overfitting: LASSO selects portfolios, ridge regularizes variance, and mean shrinkage reduces expected-return noise.
- The empirical design uses 1964-2016 CRSP/Compustat data split into training, validation, and testing periods.
- AP Trees outperform triple sorts across 36 three-characteristic cross-sections.
- Ten AP Tree portfolios often retain most of the information of larger AP Tree sets.
- Interaction nodes are crucial: removing them cuts performance substantially.
- AP Trees outperform off-the-shelf ML prediction portfolios because they target SDF spanning, not pure return prediction.
- The AP Tree advantage survives small-cap robustness checks.
- In 10-characteristic settings, AP Trees outperform deciles and large collections of double sorts with fewer portfolios.

7 Conclusion

7.1 Core conclusion

Bryzgalova, Pelger, and Zhu show that the construction of test assets is itself an asset-pricing problem. A cross-section of portfolios is useful only if it spans the SDF projection induced by the underlying characteristics. AP Trees construct such cross-sections by combining interpretable conditional sorts with robust global pruning.

7.2 Economic intuition

Characteristics affect expected returns and covariances in nonlinear and interactive ways. A fixed grid of single, double, or triple sorts cannot easily capture this structure without exploding in dimension. AP Trees

adaptively choose where to split and where to stop. Parent nodes are kept when finer splits are redundant; child nodes are kept when they improve SDF spanning.

7.3 Takeaway

The paper's main message is that researchers should not treat traditional sorted portfolios as neutral test assets. Test assets can omit important pricing information. AP Trees offer a low-dimensional, interpretable, and out-of-sample robust way to build cross-sections that are closer to the conditional investment opportunity set of individual stocks.